

Bunching Below the Line: Missing Mass Above the EU AI Act’s 10^{25} -FLOP Threshold

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analysis pass of record: 2026-07, fully deterministic (no RNG)

Abstract

After the EU AI Act entered into force on 1 August 2024, the density of new AI models just *above* the Act’s 10^{25} -FLOP systemic-risk presumption line collapses relative to just below it. In Epoch AI’s compute panel (719 models with training-compute estimates, January 2023 to June 2026), a McCrary-style binned-Poisson density test at the statutory cutoff $\log_{10} \text{FLOP} = 25.0$ gives $\theta = -1.334$ (SE 0.878, $p = 0.129$) in the primary ± 0.5 -dex window and is negative in all 15 window $\times$ bin specifications (10 of 15 with $p < 0.05$); the pre-Act period is never significant (0 of 15). A Fisher exact below/above $\times$ pre/post contrast puts the shift at odds ratio 0.173 (95% CI [0.048, 0.619], $p = 0.00715$): at the pre-Act above:below ratio, 28.9 post-Act models are expected in (25.0, 25.5] and 5 are observed — an 82.7% deficit. The deficit is specific to the statutory line (placebo thresholds at 24.5 and 25.5 are null or opposite-signed, with mass piling up just below 25.0), specific to the post-Act period (a placebo split date inside the pre-Act sample gives OR 0.90, $p = 1.0$), and correctly timed: the above-line count collapses from 2025H1, after entry into force and before the August 2025 applicability date for general-purpose AI obligations. Two results are nulls and are stated as nulls: the preferred narrow-window pre/post interaction in the discontinuity is $p = 0.729$, significant only at the widest bandwidth, and the pre-Act density test never rejects. With only 5–13 post-Act models above the line, and an Act-induced *disclosure* response that would mimic bunching exactly, we grade this as moderate evidence of bunching — behavioral or reporting — at the EU AI Act’s compute threshold, not proof of a training-compute response.

1 Introduction

Article 51(2) of the EU AI Act [2] presumes that a general-purpose AI model poses “systemic risk” when the cumulative compute used for its training exceeds 10^{25} floating-point operations, attaching notification, evaluation, and risk-mitigation obligations to models above the line. The Act entered into force on 1 August 2024 and its general-purpose AI obligations became applicable on 2 August 2025. A bright-line quantity threshold of this kind is, in the language of public economics, a *notch*: crossing it discretely changes the regulatory burden, so agents with values just above the cutoff have an incentive to locate just below it.

A large empirical literature shows that agents do exactly this. [10] developed bunching estimation at kinks in the US income-tax schedule; [7] extended it to notches, showing that a discrete

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jump in liability produces excess mass just below the cutoff and *missing mass* just above it; [6] surveys the method and its spread beyond taxation to regulation and private-sector prices. The econometric diagnostic for such sorting at a cutoff is the density discontinuity test of [9]: if agents manipulate the running variable, its density jumps at the threshold.

Whether training compute is a good handle for AI governance is actively debated. [11] argue that compute is uniquely governable — detectable, quantifiable, and produced by a concentrated supply chain — and [3] defend training-compute thresholds, including the EU’s 10^{25} -FLOP line, as the most workable trigger for regulatory oversight. [5] counters that compute thresholds are shortsighted and invite exactly the strategic responses studied in the bunching literature. On the empirical side, [8] project model counts above the EU threshold *forward* from pre-Act scaling trends, forecasting 103–306 models above 10^{25} FLOP by 2028 — a superlinear increase. To our knowledge, no published estimate tests whether the observed post-Act density of models near the line already departs from that trend. This paper supplies that test: a McCrary-style density discontinuity estimate at the statutory cutoff, before versus after the Act’s entry into force, with threshold, date, and timing placebos.

2 Data

The input panel is built from Epoch AI’s *Data on AI Models* database [1], keeping every model with a non-null positive training-compute estimate and a release date from 2 January 2023 to 4 June 2026: 719 models with columns model, date, and $\log_{10}$ training FLOP. Event dates are the Act’s entry into force (2024-08-01) and the applicability of general-purpose AI obligations (2025-08-02). The pre-Act period is 2023-01 to 2024-07 ($n = 21$ models within ± 0.5 dex of the cutoff) and the post-Act period is 2024-08 onward ($n = 52$ in-window). Compute values for recent frontier models are largely Epoch estimates rather than disclosures, and models without any compute estimate ($\approx 2,800$ of $\approx 3,500$ database rows) are necessarily excluded — a selection channel we return to in Section 5. There is no value heaping near the cutoff: at most 3 duplicate $\log_{10}$ -compute values occur within $[24.5, 25.5]$, so the results below are not an artifact of rounded compute estimates.

3 Design and Methods

Density discontinuity at the statutory line. The main estimate is the `natex` toolkit’s [4] McCrary-style [9] binned-Poisson density test, run directly at the statutory cutoff: model counts in equal-width bins of the signed distance $\log_{10} \text{FLOP} - 25.0$ are fit by a Poisson regression with a linear trend in distance and a jump term at zero; θ is the jump in log density at the cutoff, so $\theta < 0$ means a deficit just above the line — avoidance bunching below it. The primary specification uses a ± 0.5 -dex window and 10 bins; sensitivity covers the full grid $h \in \{0.5, 0.75, 1.0\} \times \text{bins} \in \{8, 10, 12, 16, 20\}$ (15 specifications per period). Because the cutoff is fixed by statute rather than searched for, the frozen-geometry caveat that applies to `natex`’s post-discovery density checks does not bite here.

Contrasts and placebos. The pre/post change in bunching is measured two ways: a Fisher exact test on the 2×2 table of below/above counts in the ± 0.5 -dex window by period, and a θ -difference (interaction) z -test between the period-specific density jumps. Specificity is probed with placebo thresholds at 24.5 and 25.5 and a placebo split date of 2023-08-01, which places both “periods” inside the pre-Act sample.

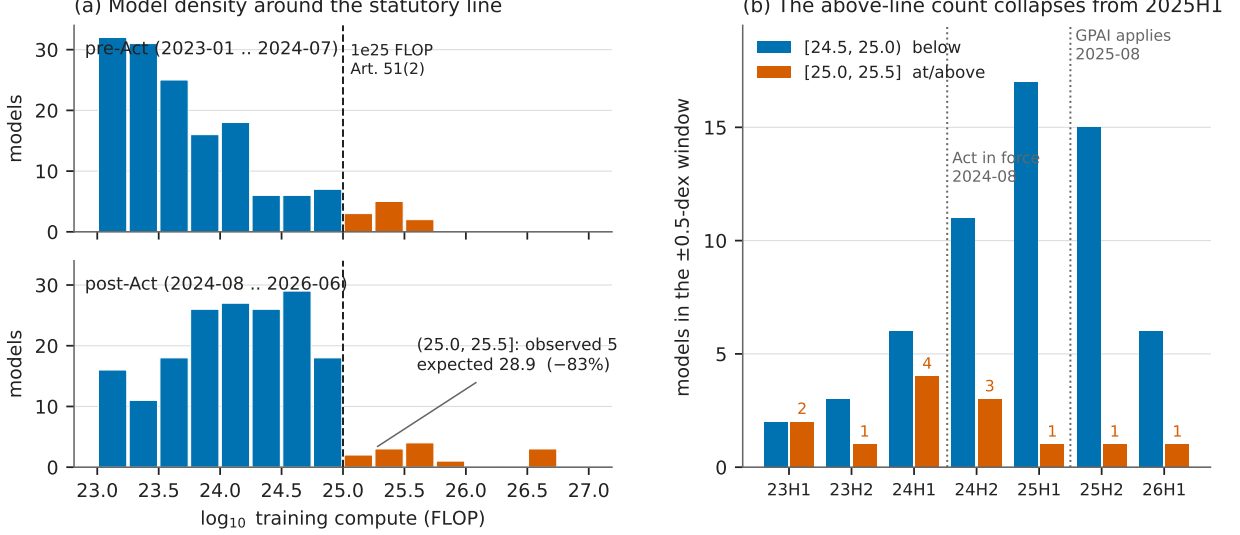


Figure 1: (a) Model counts in 0.25-dex bins of $\log_{10}$ training compute, pre-Act (top) versus post-Act (bottom), colored by side of the statutory 10^{25} -FLOP line (dashed). Post-Act, mass accumulates just below the line while the first half-dex above holds 5 models against 28.9 expected at the pre-Act ratio; the frontier jumps far above the notch. (b) Below/above counts per half-year in the ± 0.5 -dex window: the above-line count (labeled) collapses from 2025H1, after entry into force and before GPAI applicability, while the just-below count keeps growing.

Honest inference. Standard errors for θ are Wald standard errors from the binned Poisson fit and can be optimistic when the underlying density is strongly curved within the window; at the widest bandwidth ($h = 1.0$) curvature of the steeply falling compute density can leak into θ , which is why the narrow-window specification is primary and the wide-window significance is discounted below. All counts, tests, and figures derive deterministically from the input panel: there is no random number generation anywhere in the pipeline.

4 Results

A post-Act deficit just above the line. Post-Act (2024-08-01 onward), the primary density test at $\log_{10} = 25.0$ gives $\theta = -1.334$ (SE 0.878, $p = 0.129$) with $n = 52$ in-window models split 47 below / 5 above. Across the full sensitivity grid θ is negative in 15 of 15 specifications (range -1.249 to -2.566) and significant at the 5% level in 10 of 15 — all $h = 0.75$ and $h = 1.0$ specifications, with significant p -values from 0.0010 to 0.0143. The pre-Act period shows no such pattern: $\theta = -0.880$ (SE 0.973, $p = 0.366$), $n = 21$ split 13/8, and 0 of 15 sensitivity specifications reject (θ from -1.173 to $+0.494$, minimum $p = 0.243$).

The pre/post contrast. The Fisher exact test on below/above $\times$ pre/post counts in the ± 0.5 -dex window — table $[[13, 8], [47, 5]]$ — gives OR = 0.173 (95% CI $[0.048, 0.619]$, $p = 0.00715$). At the pre-Act above:below ratio, 28.9 post-Act models are expected in $(25.0, 25.5]$; 5 are observed, so roughly 24 models are “missing” — an 82.7% deficit. The formal interaction between the period-specific density jumps is a null at the primary specification and is stated as one: $\Delta\theta = -0.454$ (SE 1.311, $z = -0.346$, $p = 0.729$). Across the grid the interaction is always negative at $h \geq 0.75$ but

Table 1: Headline estimates. Density tests: binned-Poisson jump θ in log density at the statutory cutoff $\log_{10} \text{FLOP} = 25.0$; the primary window is ± 0.5 dex with 10 bins, and the sensitivity grid crosses $h \in \{0.5, 0.75, 1.0\}$ with $\{8, 10, 12, 16, 20\}$ bins. Fisher tests use below/above counts in the ± 0.5 -dex window by period.

Quantity	Estimate	SE / CI	p	notes
<i>Panel A: density jump at the statutory line</i>				
θ , pre-Act (2023-01..2024-07)	−0.880	0.973	0.366	$n = 21, 13/8$; 0/15 specs $p < 0.05$
θ , post-Act (2024-08..)	−1.334	0.878	0.129	$n = 52, 47/5$; 10/15 specs $p < 0.05$
$\Delta\theta$ (post − pre)	−0.454	1.311	0.729	<i>null at primary spec</i>
<i>Panel B: Fisher exact below/above $\times$ pre/post</i>				
threshold 25.0, Act date	OR 0.173	[0.048, 0.619]	0.00715	[[13, 8], [47, 5]]; −82.7%
placebo threshold 24.5	OR 1.637	—	0.248	post-GPAI: OR 2.510, $p = 0.0386$
placebo threshold 25.5	OR 4.00	—	0.350	opposite direction
placebo date 2023-08-01	OR 0.90	—	1.0	both halves pre-Act; no effect

Notes: $\theta < 0$ is a density deficit just above the cutoff. SEs are Wald standard errors from the binned Poisson fit. Significant sensitivity p -values for the post-Act θ range from 0.0010 to 0.0143 (all $h = 0.75$ and $h = 1.0$ specifications); the interaction is significant only at $h = 1.0$ with ≥ 12 bins ($p = 0.0049$ – 0.0342). Placebo-threshold odds ratios use the same ± 0.5 -dex construction centered on the placebo line.

significant only at $h = 1.0$ with 12/16/20 bins ($p = 0.0249/0.0049/0.0342$; $h = 1.0$ with 10 bins gives $p = 0.0639$), where curvature leakage is a concern (Section 3). The pre-period, with only 21–47 in-window models, is too small to pin down the counterfactual discontinuity.

Placebos. The deficit is specific to the statutory line and period. At placebo threshold 24.5 the same pre/post Fisher contrast gives $\text{OR} = 1.637$, $p = 0.248$; restricted to the post-GPAI period it gives $\text{OR} = 2.510$, $p = 0.0386$ with the *opposite* sign — more mass in $[24.5, 25.0)$ than $[24.0, 24.5)$, i.e. mass piling up just below 25.0, consistent with bunching at the line and inconsistent with a generic high-compute slowdown. At placebo threshold 25.5: $\text{OR} = 4.00$, $p = 0.350$, direction opposite to the 25.0 effect. The placebo split date 2023-08-01, with both halves pre-Act, gives $\text{OR} = 0.90$, $p = 1.0$ at the true threshold — no pseudo-effect.

Timing. Below:above counts per half-year in the ± 0.5 -dex window are 2023H1 2:2, 2023H2 3:1, 2024H1 6:4, 2024H2 11:3, then 2025H1 17:1, 2025H2 15:1, 2026H1 6:1 (Figure 1b). The collapse begins in 2025H1 — after entry into force (August 2024), before the obligations became applicable (August 2025) — consistent with anticipatory compliance: the 10^{25} presumption was public from July 2024 and attaches to models placed on the market from August 2025. By year, $[25.0, 25.5)$ holds 3, 7, 2, 1 models in 2023–2026 while $[24.5, 25.0)$ grows 5, 17, 32, 6. Secular compute growth biases *against* the finding — later cohorts should put more, not less, mass above any fixed line, and do at 24.5. Meanwhile 13 post-Act models sit at or above 10^{25} at some height, with the frontier jumping far above the notch (Grok 3 at $\log_{10} = 26.54$, GPT-4.5 at 26.58, Llama 4 Behemoth preview at 25.71, Grok 4 at 26.70, GPT-5 at 25.82) — the classic bunching signature in which a notch distorts only marginal decisions [7].

5 Caveats and Conclusion

Three caveats bound the claims. First, *small n*: only 5 models sit in $(25.0, 25.5]$ and 13 at any height above 10^{25} post-Act; the preferred narrow-window pre/post interaction is $p = 0.729$, and the interaction is significant only at $h = 1.0$ with ≥ 12 bins (p from 0.0049 to 0.0342), where curvature of the steeply falling density can leak into θ . Second, *measurement, not behavior*: post-2024 frontier compute values are largely Epoch estimates, and models without a compute estimate are excluded (719 of $\approx 3,500$ rows have compute). If the Act suppresses *disclosure* of parameters and token counts near the line, the identical density gap appears with no training-compute response at all; distinguishing a compliance response from a reporting response requires disclosure-side data and is the designated follow-up. Third, a *confounded trend break*: the deficit’s onset (2025H1) coincides with the industry shift toward inference-time scaling and distilled or mixture-of-experts models that followed the late-2024 o1-style reasoning models, which independently reduced demand for 10^{25} -plus pre-training runs among non-frontier labs — though this confound predicts a broad high-compute slowdown, not a deficit specific to the statutory line with mass accumulating immediately below it.

The conclusion is graded accordingly. The post-Act deficit just above the EU AI Act’s 10^{25} -FLOP line is sign-consistent in all 15 density specifications, large (Fisher $p = 0.00715$, 82.7% missing mass), specific to the statutory threshold and period, and correctly timed, with the marginal-crossers-vanish, frontier-jumps-far-above signature that notch responses produce. But with 5–13 post-Act models above the line, a null preferred narrow-window interaction, and an Act-induced disclosure response that would mimic bunching exactly, it is moderate evidence of bunching — behavioral or reporting — rather than proof of a training-compute response. Either reading is informative for the compute-threshold debate [3, 5]: the forward projections of [8] assumed threshold-blind scaling, and the data just above the line already look threshold-aware.

Reproducibility. All estimates derive deterministically from the public Epoch AI panel; no dataset files are committed. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

References

- [1] Epoch AI (2026). Data on AI models. <https://epoch.ai/data/ai-models> (compute panel extracted June 2026).
- [2] European Union (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). *Official Journal of the European Union*, L series, 12 July 2024.
- [3] Heim, L., and Koessler, L. (2024). Training compute thresholds: Features and functions in AI regulation. arXiv:2405.10799.
- [4] Hillebrandt, H. (2026). *natex: automated natural-experiment discovery and estimation*. Software. <https://github.com/HaukeHillebrandt/natex>
- [5] Hooker, S. (2024). On the limitations of compute thresholds as a governance strategy. arXiv:2407.05694.
- [6] Kleven, H. J. (2016). Bunching. *Annual Review of Economics*, 8, 435–464.
- [7] Kleven, H. J., and Waseem, M. (2013). Using notches to uncover optimization frictions and structural elasticities: Theory and evidence from Pakistan. *Quarterly Journal of Economics*, 128(2), 669–723.
- [8] Kumar, I., and Manning, S. (2025). Trends in frontier AI model count: A forecast to 2028. arXiv:2504.16138.
- [9] McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics*, 142(2), 698–714.
- [10] Saez, E. (2010). Do taxpayers bunch at kink points? *American Economic Journal: Economic Policy*, 2(3), 180–212.
- [11] Sastry, G., Heim, L., Belfield, H., Anderljung, M., Brundage, M., Hazell, J., O’Keefe, C., Hadfield, G. K., et al. (2024). Computing power and the governance of artificial intelligence. arXiv:2402.08797.